

Where the neurologists are not: measuring the gap between neurological disease burden and the capacity to treat it

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Significance statement. Specialists per country is the statistic normally used to describe where medical capacity sits, and it answers the wrong question. Combining three World Health Organization sources across 52 countries, this study shows that the countries holding the largest share of the world’s stroke burden hold the smallest share of its neurologists, by a factor of roughly thirty-four. It then shows that national counts overstate what most residents can reach, because the poorer a country is, the more completely its specialists concentrate in the capital. In low-income countries the specialist density facing a rural resident is not merely low but exactly zero, which makes the usual language of ratios inapplicable.

) contains a list environment (quote, quotation, theorem, definition, enumerate, itemize...), the line after the list may have some extra indentation. If this is the case, add to the end of the list environment. Neurological disorders are among the largest contributors to global disability [1], and the workforce able to diagnose and manage them is among the most unevenly distributed of any medical specialty. Demand is also rising fastest where capacity is thinnest: dementia prevalence is forecast to almost triple by 2050, with the steepest growth in low- and middle-income regions [2]. This paper asks a distributional question: are the countries carrying the heaviest neurological disease burden the ones with the capacity to treat it?

Related work

The question is not new, and this is not the first analysis of neurological workforce scarcity. The WHO and World Federation of Neurology Atlas has documented that scarcity across two editions and established that resources are severely limited in low-income countries [6]. Feigin and colleagues set out the case for translating burden evidence into workforce policy [1], and comparative health-system statistics show specialist supply concentrated in high-income settings, with rural residents facing additional access barriers even inside wealthy countries [3].

What the Atlas does not do, and what this analysis adds, is link capacity to *need*. The Atlas reports medians within income groups and WHO regions, presents no country-level workforce estimates, and includes no burden measure at all. It can therefore say that low-income countries have fewer neurologists, but not whether the countries with the least capacity are the ones carrying the heaviest burden, which is the question that matters for priority-setting. Grouped medians also suppress within-group variation entirely, so a low-income country with functioning services and one with none are indistinguishable in the published tables. This paper extends that literature in three ways: it joins capacity to a burden measure at the country level, it reports a distributional statistic rather than a central tendency, and it separates how much capacity exists from where inside a country that capacity sits.

Data

Sources. Three WHO products and one survey inventory are used. (i) The *Global Dementia Observatory* [5] indicator GDO_q6x1_2 gives neurologists per 100,000 population, self-reported by ministries of health. (ii) GDO indicator GDO_q8x3_1 gives the furthest geographic level at which dementia diagnostic services are available, on an ordered three-level scale. (iii) WHO *Global Health Estimates* series SA.0000001689 [4] gives age-standardized stroke DALYs per 100,000. (iv) The WHO/WFN *Neurology Atlas*, second edition [6], supplies the country to WHO-region to World-Bank-income-group crosswalk from Annex 1 and the aggregate

workforce gradients from Tables 2–3 and Figures 11, 12 and 16. A DHS survey inventory is used only to describe coverage for a possible subnational extension.

Time window. Capacity and service reach are 2017, the fielding year of the second Atlas and the GDO round used. Burden is **2004**, which is the only round of the WHO stroke DALY series exposed by the GHQ API at the time of writing. The thirteen-year offset is the single largest weakness in the analysis and is the reason RQ1 is framed as describing a gap rather than estimating one.

Unit of analysis. In the original sources the unit differs by product: GDO records are one row per country per indicator per round; GHE records are one row per country per cause per age-standardization per year; the Atlas Annex is one row per responding Member State. The unit of analysis *used here* is the **country**. Every source is collapsed to one row per ISO3 code before merging, and no country appears twice.

Missingness in key fields. The panel has 52 countries. Neurologist density is missing for **33 of 52** (63%), service reach for 5 (10%), income group for 13 (25%), and stroke burden for none. WHO encodes absence as the strings “Not available”, “Not applicable”, “No data” and “Not reported” rather than as empty cells; these are converted to missing in `code/03_clean_merge.py` (lines 19–20) before any numeric coercion, so that a placeholder string cannot silently become a zero. Missingness is not random: countries that report a workforce count plausibly have stronger health information systems, which biases the sample toward better-resourced countries and therefore *understates* the true gap.

Limitations of the sources. All WHO instruments here are ministry questionnaires, self-reported and unaudited, with no full-time-equivalent adjustment, so a neurologist working one clinic day a week counts the same as one working five. Stroke stands in for neurological burden generally. Measured burden itself depends on diagnostic capacity, which is the exposure of interest, so the two are entangled by construction; this is addressed by bounding rather than by adjustment.

Methods

Cleaning. `code/03_clean_merge.py` performs every cleaning step. WHO placeholder strings are mapped to missing (line 20); the ordered service-reach question is mapped from text to an integer scale via `ACC_SCALE` in `utils.py` (line 36); and two binary indicators are derived from that scale, `beyond_capital` for levels ≥ 2 and `reaches_rural` for level 3 (lines 42–43).

Merging. Capacity and service reach are combined with an **outer join on ISO3**, deliberately, so that a country reporting one but not the other is retained rather than dropped; this yields 52 rows. Burden and the Atlas crosswalk are then attached with **left joins on ISO3**, preserving those 52 rows. All matching is **exact on ISO3 country code**, not fuzzy on country name, which avoids the standard failure mode where “Republic of Korea” and “Korea, Rep.” fail to match. The script prints row counts before and after every join, and the printed counts appear in `output/log_03_clean_merge.txt`.

Variable definitions. The **burden-to-capacity share ratio** for country i is

$$R_i = \frac{b_i / \sum_j b_j}{c_i / \sum_j c_j} \quad (1)$$

where b_i is the age-standardized stroke DALY rate and c_i is neurologist density. Both terms are shares, so R_i is invariant to the units of either input. $R_i = 1$ means country i holds exactly as large a share of the sample’s burden as of its workforce; $R_i = 13.1$ means the same staff are stretched over thirteen times the need. It is undefined at $c_i = 0$, and those countries are reported separately rather than dropped.

Table 1: The analysis sample. Row counts are after the merges described in Methods; every source is 2017 except stroke burden, which is 2004. The two analysis samples differ because reporting a workforce count and reporting a service-reach level are separate questionnaire items, answered by overlapping but non-identical sets of countries.

Measure	n
Countries in merged panel	52
with neurologist density (WHO GDO, 2017)	19
with service-reach level (WHO GDO, 2017)	47
with stroke DALY rate (WHO GHE, 2004)	52
with World Bank income group (Atlas, 2017)	39
RQ1 analysis sample: capacity and burden	19
RQ3 analysis sample: reach and income group	35

Countries reporting zero neurologists are retained in the panel but cannot be placed on the ratio scale of Eq. 1.

The **Urban Concentration Index** and **effective rural density** are

$$\text{UCI}_g = \frac{p_g^{\text{cap}} - p_g^{\text{rur}}}{100} \quad (2)$$

$$D_g^{\text{eff}} = m_g \cdot \frac{p_g^{\text{rur}}}{100} \quad (3)$$

where p_g^{cap} and p_g^{rur} are the percentages of countries in income group g reporting neurologists in the capital and in rural areas, and m_g is the group’s median density. UCI is bounded $[0, 1]$; a value of 1 means every country has capital coverage and none has rural coverage. D_g^{eff} rescales national density by the share of countries where a rural resident could reach anyone at all.

Inequality is measured by the **Gini coefficient** of neurologist density, computed by the sorted-index formula in `utils.py` (line 214) and independently by trapezoidal integration under the Lorenz curve (line 233). The two agree to 1.1×10^{-16} , which is a check on the implementation rather than a result.

Inference, and why no country-level regression is reported. This is a **descriptive** analysis. There is no source of causal identification: no randomization, no instrument, no policy discontinuity, and national income drives both capacity and measured burden. No causal effect is estimated or implied.

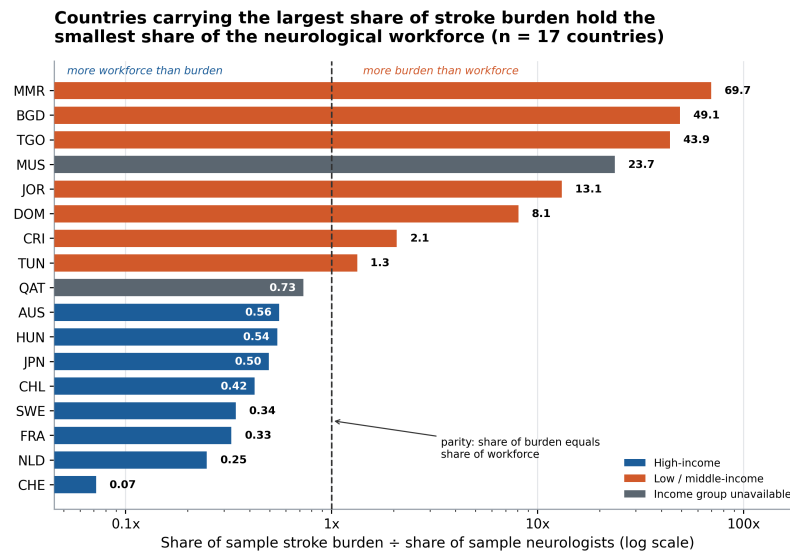
With 19 to 35 observations, a fitted country-level slope would be badly underpowered, so no slope is reported as a headline result. Countries are binned by World Bank income group and the groups compared directly, with a Welch t -test, a Mann-Whitney U , a 20,000-draw permutation test that assumes nothing about distributional shape, a one-way ANOVA and Kruskal-Wallis across three bins, a chi-square on the reach-by-income contingency table, and a logistic regression of rural reach on high income. A regression *is* fitted in `code/07_regression_and_interactions.py`, but only to demonstrate two solvers agreeing (normal equations against gradient descent, to 4.4×10^{-16}) and to test an interaction the sample cannot resolve.

Results

The analysis sample. Table 1 characterizes it. Of 52 countries in the merged panel, 19 support the RQ1 comparison and 35 the RQ3 comparison; the difference is entirely driven by which WHO indicators a country chose to answer.

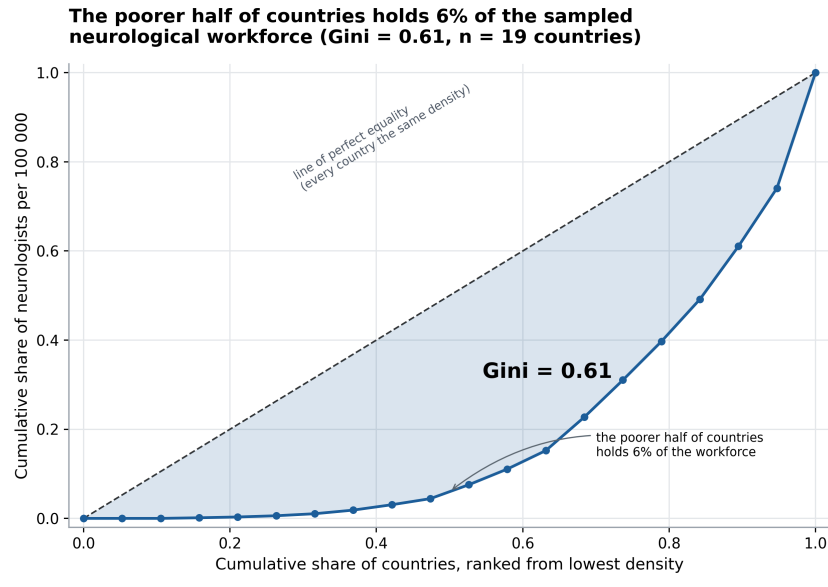
Capacity is not aligned with burden

On the share ratio, the median high-income country sits at **0.38** and the median low- or middle-income country at **13.12**, a factor of **34.3**. Myanmar reaches 69.7, Bangladesh 49.1, Togo 43.9. At the other end Switzerland sits at 0.07. Eswatini and Fiji report *zero* neurologists and cannot be placed on the scale at all, while carrying stroke DALY rates of 1,536 and 994 per 100,000. Figure 1 shows the full ordering.



A ratio above 1.0 means the country holds a larger share of the sample's stroke burden than of its neurologists; below 1.0 the reverse. The axis is logarithmic because the ratio spans three orders of magnitude, which also places parity at the center. Capacity is WHO Global Dementia Observatory neurologist density per 100 000 (2017); burden is WHO Global Health Estimates age-standardized stroke DALYs per 100 000 (2004). Eswatini and Fiji report zero neurologists and cannot be placed on a ratio scale at all; their stroke DALY rates are 1536 and 994 per 100 000.

Figure 1: Burden-to-capacity share ratio by country, logarithmic axis. A ratio above 1.0 means the country holds a larger share of the sample's stroke burden than of its neurologists; below 1.0 the reverse. The axis is logarithmic because the ratio spans three orders of magnitude, which also places parity at the center. Capacity is WHO Global Dementia Observatory neurologist density per 100,000 (2017); burden is WHO Global Health Estimates age-standardized stroke DALYs per 100,000 (2004). $n = 17$ countries placeable on the scale; Eswatini and Fiji report zero neurologists and are excluded from the axis but discussed in the text. Median high-income 0.38, median low/middle-income 13.12, a factor of 34.3.



Lorenz curve of WHO Global Dementia Observatory neurologist density per 100 000 population (2017). Countries are ranked from lowest to highest density along the horizontal axis; the vertical axis is the cumulative share of the sampled workforce they hold. A Gini of 0 would place the curve on the dashed diagonal, a Gini of 1 in the bottom-right corner. The shaded area is half the Gini by construction. Highest density: CHE at 13.1 per 100 000. Lowest non-zero: MMR at 0.07. Two further countries report exactly zero and sit on the flat segment at the left.

Figure 2: Lorenz curve of neurologist density across the 19 countries reporting a workforce count. The horizontal axis is the cumulative share of countries, ordered from lowest density to highest; the vertical axis is the cumulative share of the sampled workforce they hold. Perfect equality would place the curve on the diagonal, and the shaded area between the two is half the Gini coefficient by construction. The poorest half of countries holds 6% of the workforce; the Gini is 0.61 (95% bootstrap CI 0.43–0.73). Source: WHO/WFN Neurology Atlas, 2nd edition (2017).

Capacity is distributed very unequally

The Gini coefficient of neurologist density across the 19 reporting countries is **0.61**, with a 95% bootstrap interval of **[0.43, 0.73]**: high inequality, imprecisely estimated. The poorer half of countries holds about **6%** of the sampled workforce. Inequality is greater between income groups than within them, with a Gini of 0.27 within high-income countries and 0.49 within low- and middle-income countries. The focal contrast is Switzerland at 13.10 per 100,000 against Myanmar at 0.07, a **187-fold** difference. Dropping any single country moves the Gini by at most 0.06, so no one reporter carries the result.

The gap differs systematically between country types

Table 2 reports the binned comparisons. Neurologist density differs between income bins on every test applied, including the distribution-free permutation test. The most striking descriptive result is that dementia diagnostic services reach rural areas in 77% of high-income countries and in **none** of the four low or lower-middle income countries sampled.

The gradient steepens with rarity

Across the 114 Atlas respondents, the high-to-low income ratio is 71-fold for total neurological workforce, 158-fold for adult neurologists, 62-fold for neurosurgeons and **195-fold** for child neurologists. The scarcer the cadre, the more concentrated it is in rich countries.

Table 2: Binned hypothesis tests. Country-level regressions are not fitted as headline results because 19 to 35 observations cannot support them, so countries are binned by World Bank income group and the groups compared directly. Rank-based and permutation tests are reported alongside the parametric ones because the chi-square has six of nine cells with expected counts below five.

Comparison	Test	Result
Density, high vs low/middle	Welch t	$t = 4.29, P = 0.0031$
same	Mann-Whitney U	$P = 0.0003$
same	Permutation (20,000)	$P = 0.0003$
Reach across three bins	One-way ANOVA	$F = 5.74, P = 0.0074$
same	Kruskal-Wallis	$H = 9.44, P = 0.0089$
Reach $\times$ income bin	Chi-square	$\chi^2 = 11.81, P = 0.0188$
Rural reach $\sim$ high income	Logistic	OR = 7.6, $P = 0.0098$

All tests are two-sided. The permutation test shuffles income-group labels 20,000 times and assumes nothing about distributional shape.

National density overstates access

Figure 3 reports both indices. UCI falls monotonically from 0.95 in low-income countries through 0.83 and 0.78 to 0.52 in high-income ones. Effective rural density rises across the same groups from **exactly zero** through 0.24 and 0.59 to 3.19 per 100,000. The zero is not a rounding artifact: no low-income country in the Atlas reports a single neurologist practicing rurally. The national gap between high- and low-income countries is 71-fold; the rural gap has no denominator at all.

An interaction the sample cannot resolve

If capacity buys less alignment where services remain urban, capacity and rural reach should interact. Only **13 countries** carry both measures. The interaction coefficient is -0.004 with $P = 0.99$. That is a statement about statistical power, not evidence that no interaction exists, and the figure title in the repository says so explicitly rather than implying a null finding.

Robustness

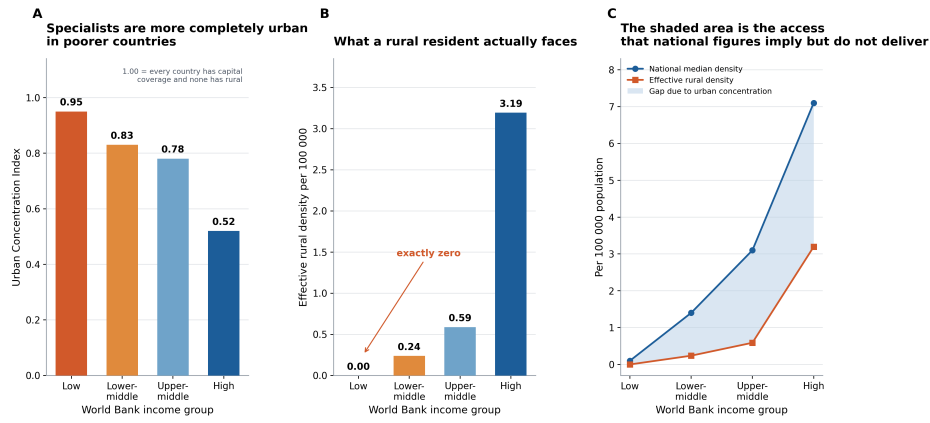
Three checks. A 5,000-draw bootstrap gives the Gini interval quoted above. Leave-one-out moves the Gini by at most 0.06 for any single country. A bounding exercise asks how much measurement error would be needed to overturn the central result: burden in low- and middle-income countries would have to be **overstated by a factor of 34**, or their capacity understated by the same factor, for the two median ratios to level. Measurement error of that magnitude is not plausible, and under-diagnosis in low-capacity settings pushes *measured* burden down rather than up, which widens the true gap rather than closing it.

Discussion

Across three metrics the finding is consistent: neurological capacity is concentrated where measured burden is lightest, and national counts overstate what most residents can reach. The policy implication is that specialists per country is the wrong statistic for priority-setting. It answers a question about supply when the operative question is about reach, and it is silent about whether a rural resident can access anyone at all.

Six limitations bound the conclusion. The thirteen-year offset between capacity and burden is the weakest joint. The samples are small and self-selected toward countries with functioning health information systems, which understates the gap. Stroke stands in for neurological burden generally. Six of nine cells in the reach-by-income table have expected counts below five. Both WHO instruments are self-reported and unaudited with no FTE adjustment. And the interaction test is underpowered at 13 countries. Nothing here is a causal estimate.

National density overstates access wherever specialists stay urban



WHO and World Federation of Neurology Atlas, 2nd edition (2017), Figures 12 and 16; 114 responding countries. Panel A: the Urban Concentration Index is the share of countries reporting specialists in the capital minus the share reporting any rural practice, divided by 100. It is bounded on [0, 1] because capital reach weakly dominates rural reach in every group observed. Panel B: effective rural density multiplies the national median by the share of countries with any rural practice, so it answers what a rural resident faces rather than what the national average is. It is exactly zero for low-income countries because no low-income country in the Atlas reports a neurologist practicing rurally, which makes the high-to-low gap undefined rather than merely large. Panel C plots both series on one axis.

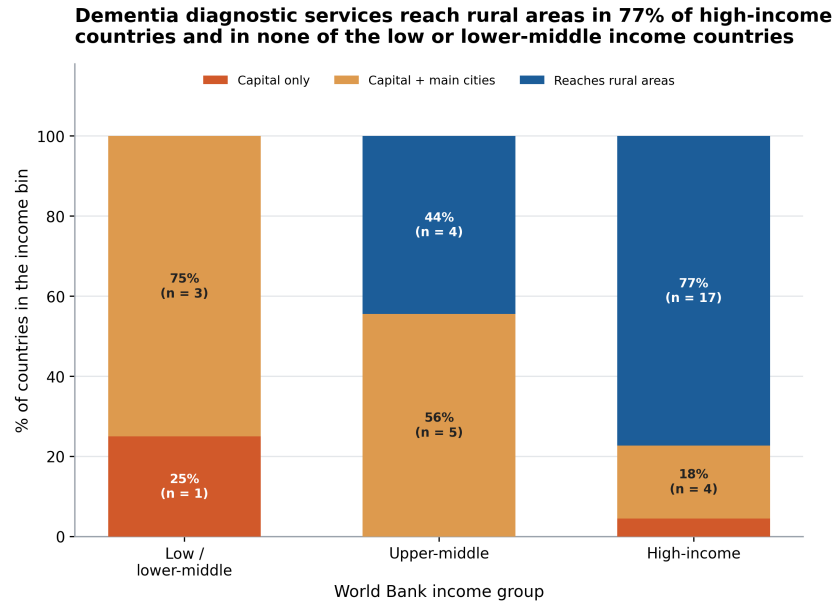
Figure 3: National density overstates access wherever specialists stay urban. Panel A: the Urban Concentration Index, the share of countries where neurologists reach the capital minus the share where they reach rural areas. Panel B: effective rural density, the national median rescaled by the share of countries with any rural practice, which is the quantity a rural resident actually faces. The low-income bar in Panel B is exactly zero, not rounded to zero. Values by income group, low to high: UCI 0.95, 0.83, 0.78, 0.52; effective rural density 0.00, 0.24, 0.59, 3.19 per 100,000. WHO/WFN Neurology Atlas, 2nd edition (2017), 114 responding countries.

Two extensions follow directly. The DHS inventory described in the repository covers 106 countries and 443 surveys, 62 with GPS coordinates, which would permit the same alignment question to be asked *within* countries rather than across them. And a burden series contemporaneous with the 2017 capacity measures would remove the study's largest weakness.

Appendix A. The remaining figures

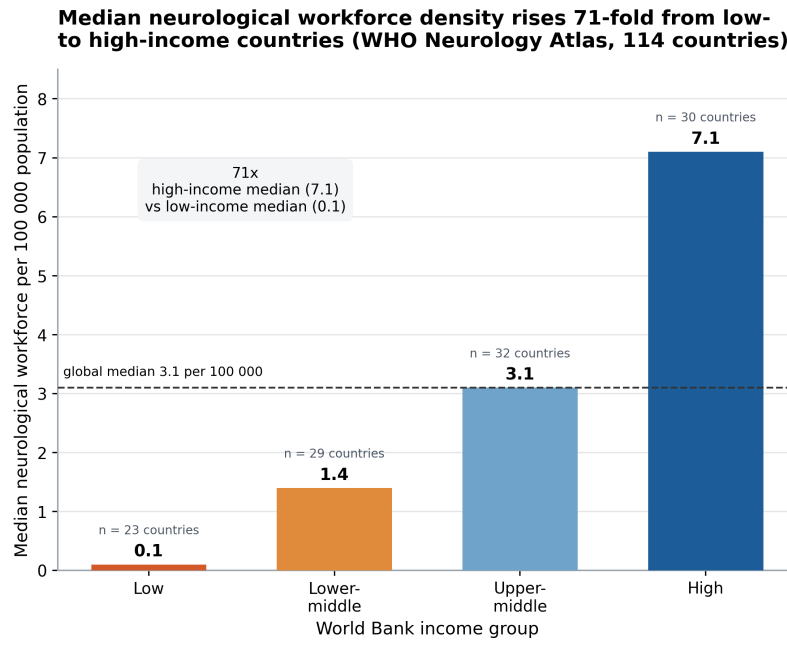
Three figures and three tables appear in the body. The pipeline produces fourteen figures and ten tables. The eleven held back are not weaker versions of those shown; they answer adjacent questions, and several exist to check the analysis rather than to say anything about neurologists. All are reproduced here with the same self-contained captions they carry in the repository. This appendix and the Agentic Analysis section that follows are excluded from the page limit.

A.1 Extending the descriptive picture



WHO Global Dementia Observatory (2017) indicator GDO_q8x3_1, cross-tabulated with World Bank income group taken from Annex 1 of the WHO Neurology Atlas. $n = 35$ countries reporting both fields. Bars sum to 100% within each income group because the question is single-choice. Chi-square = 11.81, $df = 4$, $p = 0.0188$, but 6 of 9 cells have an expected count below 5, so the rank-based and permutation tests reported in table t02 are the ones to rely on. Low and lower-middle income countries are pooled because only four low-income countries answered this item.

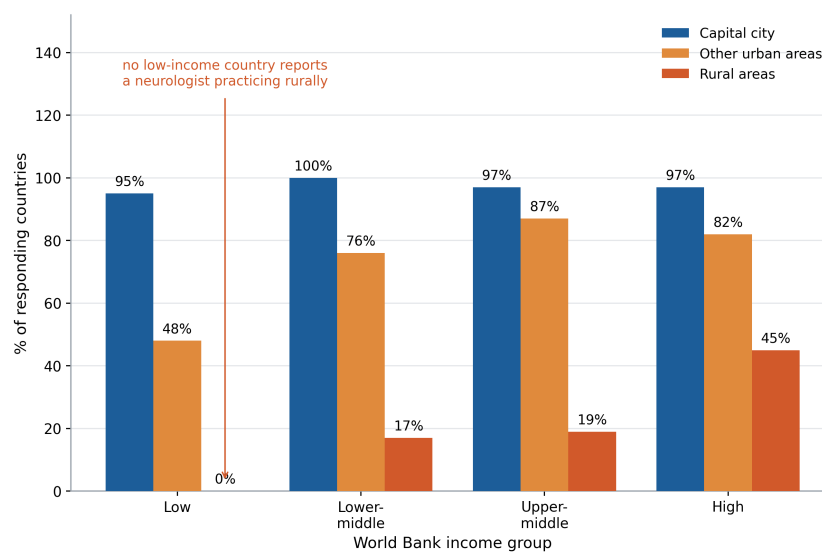
Figure 4: Dementia diagnostic service reach by World Bank income group. The ordered scale runs capital only, capital plus main cities, reaches rural areas. Rural reach occurs in 77% of high-income countries and in none of the four low or lower-middle income countries sampled. WHO Global Dementia Observatory (2017), $n = 35$ countries with both a reach level and an income group.



Median of the total neurological workforce, defined by the Atlas as adult neurologists plus neurosurgeons plus child neurologists, per 100 000 population within each World Bank income group. Source: WHO and World Federation of Neurology, Atlas: Country Resources for Neurological Disorders, 2nd edition (2017), Figure 12. The count under each bar is the number of countries answering that item. These are medians of country values and are not weighted by population, so a group median describes the typical country in the group rather than the experience of the typical person in it.

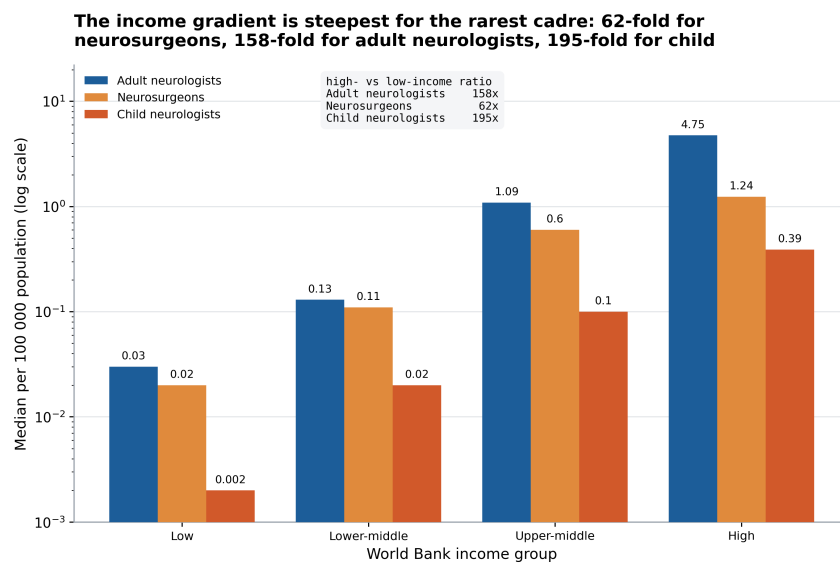
Figure 5: Median neurological workforce per 100,000 by income group across all 114 Atlas respondents, rather than the 19 with a usable country-level count. The high-to-low ratio is 71-fold. Source: WHO/WFN Neurology Atlas, 2nd edition (2017), Figure 12.

Capital-city coverage is near-universal at every income level while rural coverage falls from 45% to 0% (WHO Neurology Atlas, n = 114)



Share of responding countries reporting that neurologists practice in each setting, by World Bank income group. A country can report more than one setting, so bars within a group do not sum to 100%. Source: WHO and World Federation of Neurology Atlas, 2nd edition (2017), Figure 16; 114 responding countries. This is the aggregate counterpart to Figure 3, which measures the same gradient country by country in the smaller Global Dementia Observatory extract. The two instruments use different questions and different country sets and produce the same ordering.

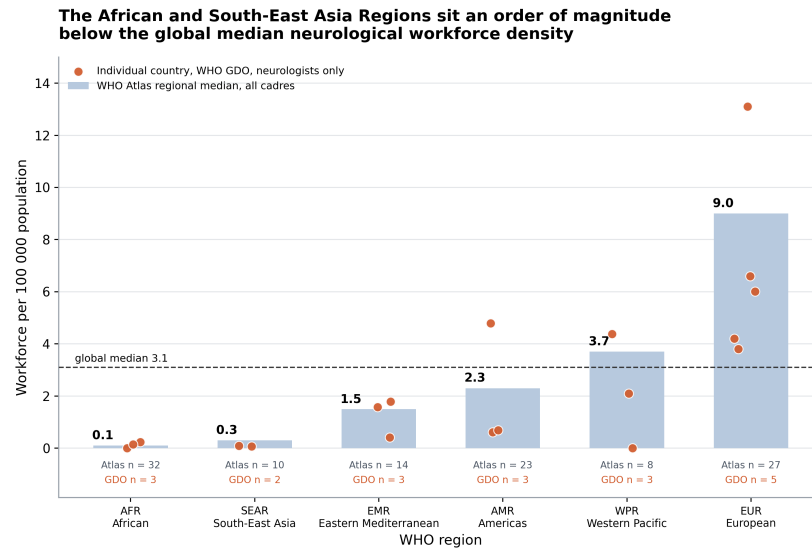
Figure 6: Where neurologists practice, by income group. Capital-city coverage is near universal at every income level; rural coverage falls from 45% to 0%. This is the aggregate counterpart to Fig. 4, measured by a different instrument on a larger country set, and it produces the same ordering. WHO/WFN Neurology Atlas (2017), Figure 16, 114 responding countries.



Median number of each cadre per 100 000 population by World Bank income group, on a logarithmic vertical axis because the values span four orders of magnitude, from 0.002 to 4.75. A logarithmic axis makes equal ratios equal distances, so the widening vertical spacing between the three series from left to right is itself the finding rather than an artifact of the scale. Source: WHO and World Federation of Neurology Atlas, 2nd edition (2017), Table 3. Response counts differ by cadre, from 93 countries for child neurologists to 114 for adult neurologists, and are reported in table t03.

Figure 7: The income gradient by clinical cadre, logarithmic axis. The scarcer the cadre, the more concentrated it is in high-income countries: 71-fold for total workforce, 158-fold for adult neurologists, 62-fold for neurosurgeons and 195-fold for child neurologists. WHO/WFN Neurology Atlas (2017), Tables 2–3.

A.2 Geography rather than economics



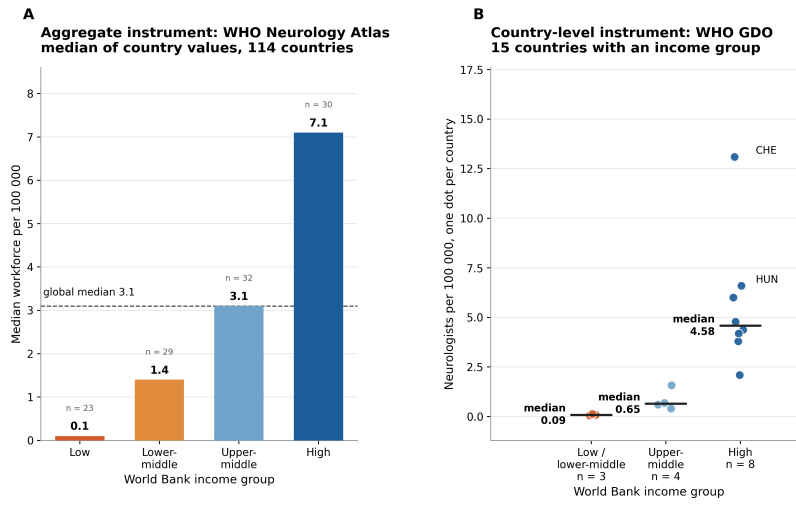
Bars are the WHO and World Federation of Neurology Atlas median of the total neurological workforce per 100 000 population within each WHO region (2017, Figure 11); the count under each bar is the number of countries answering that item. Orange dots are individual countries from the WHO Global Dementia Observatory (2017), which counts adult neurologists only and therefore sits systematically below the Atlas bar for the same region. Dots are jittered horizontally to avoid overplotting. Regions are ordered by Atlas median rather than alphabetically. The two instruments track each other closely: the GDO median for the European Region is 6.01 against the Atlas adult-neurologist median of 6.60, and for South-East Asia 0.08 against 0.10.

Figure 8: Median workforce density by WHO region with individual reporting countries overlaid, so within-region spread remains visible. Income group and region are correlated but not identical, and the overlay is what shows the difference. WHO/WFN Neurology Atlas (2017) and WHO Global Dementia Observatory (2017).

A.3 Paired instrument checks

Each panel places an Atlas aggregate beside its country-level counterpart from the Global Dementia Observatory. Two instruments collected independently, by different teams, with different questions, ordering countries the same way is weak evidence that neither is simply wrong. It is not independent confirmation, because both are ministry self-reports.

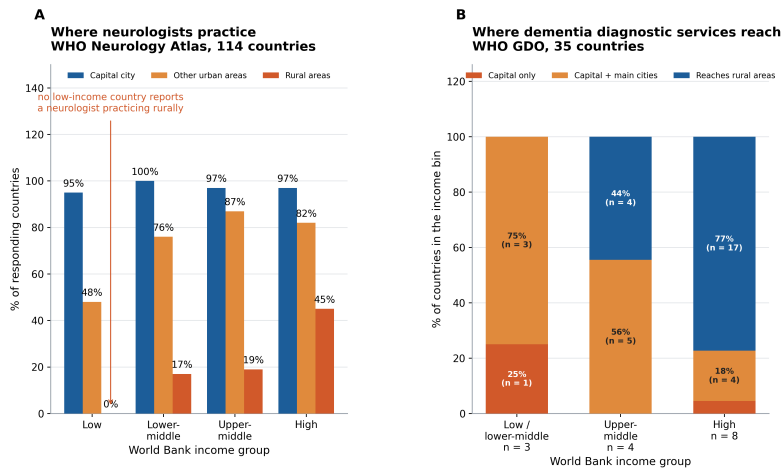
The same workforce gradient measured by two independent WHO instruments



Panel A reports the median across countries within each income group and cannot identify individual countries. Panel B reports each country separately but covers far fewer of them. The two disagree on level, because Panel A counts adult neurologists, neurosurgeons and child neurologists while Panel B counts adult neurologists only, and agree on direction and steepness. That agreement is the point: the country-level finding is not an artifact of the small sample. The three lowest-density countries in Panel B, too close together to label in place, are MMR 0.07, BGD 0.09, TGO 0.14 per 100 000. Sources: WHO and World Federation of Neurology Atlas 2nd edition (2017) Figure 12; WHO Global Dementia Observatory (2017) GDO_q6x1_2.

Figure 9: The same workforce gradient measured two ways. Panel A reports the median within each income group and cannot identify individual countries; Panel B reports each country separately but covers far fewer of them. They disagree on level, because Panel A counts three cadres and Panel B counts adult neurologists only, and agree on direction and steepness.

Two instruments, one finding: specialist care thins out before it reaches rural populations

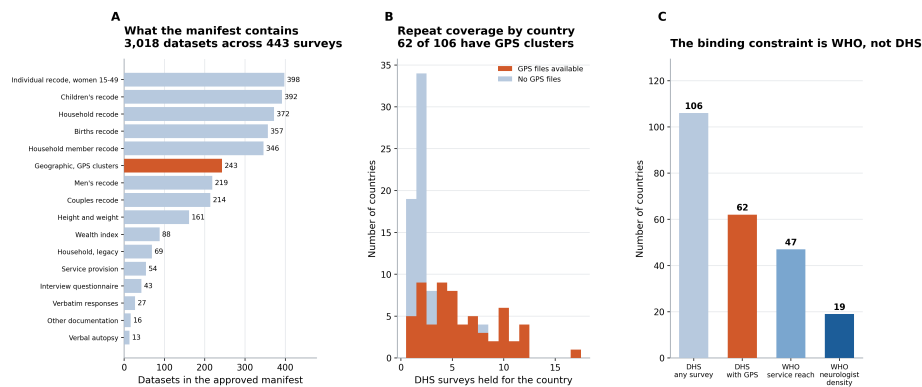


Panel A asks whether neurologists practice in a given setting at all, and a country may report several. Panel B asks how far dementia diagnostic services extend, and the answer is single-choice. These are different questions about the same underlying constraint, asked of overlapping but not identical country sets, and they produce the same ordering: rural coverage falls from 45% to 0% across income groups in Panel A, and the share of countries whose services reach rural areas falls from 77% to 0% in Panel B. Sources: WHO Atlas 2nd edition (2017) Figure 16; WHO Global Dementia Observatory (2017) GDO_q8x3_1.

Figure 10: The same rural-access gradient measured two ways, with the Atlas aggregate beside the country-level Global Dementia Observatory extract.

A.4 Groundwork and verification

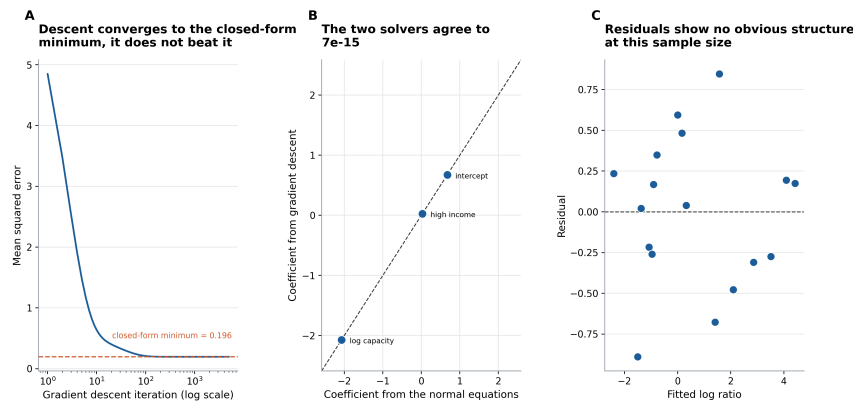
The approved DHS holdings are broad; the WHO capacity data are the limiting factor



Derived from the filenames in the approved DHS Program download manifest. No survey data were downloaded or read: the DHS filename convention CCTVFL.zip encodes country, file type and survey phase, which is enough to build a coverage inventory. GE files carry displaced GPS cluster coordinates and are the input for the distance-to-care measure planned as the next step. Panel C is the reason the country-level analysis in Figures 1 to 3 is small: DHS covers 106 countries, but only 19 of them have a WHO neurologist-density value to link them to.

Figure 11: Demographic and Health Survey coverage across the sampled countries: 3,018 datasets, 106 countries, 443 surveys, 62 with GPS coordinates. This is groundwork for asking the alignment question *within* countries rather than across them. It is not a result.

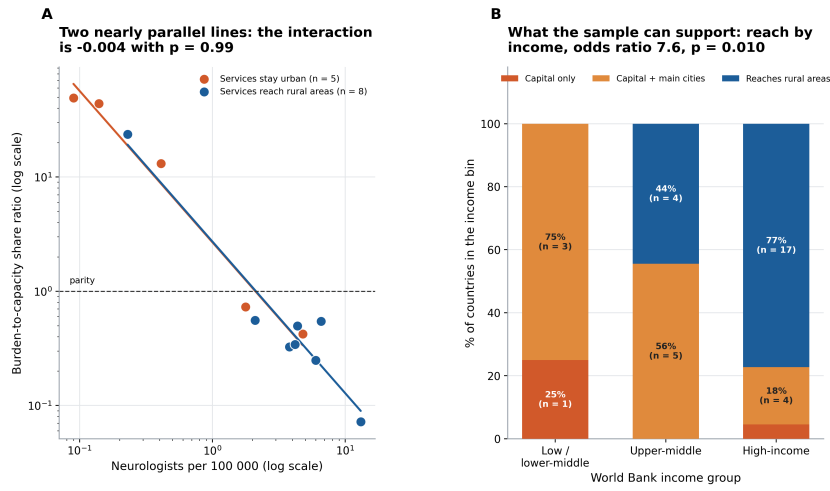
The country-level regression, solved two ways as a check



Outcome is the natural log of the burden-to-capacity share ratio for the 17 countries with both a burden and a positive capacity value. Predictors are standardized so a single learning rate suits both. Panel A: gradient descent minimizes the same mean squared error the normal equations solve exactly, so the curve approaches the dashed line from above and never crosses it. Panel B: coefficients from the two solvers plotted against each other; points on the diagonal mean they agree. Panel C: residuals against fitted values, the standard check for non-constant variance or curvature, neither of which is visible here, though 17 points is too few to see much. Capacity appears in the denominator of the outcome, so its large negative coefficient is mechanical rather than a substantive finding.

Figure 12: The two estimators of the same regression, closed-form normal equations against gradient descent, agreeing to 4.4×10^{-16} . This verifies the implementation and says nothing about neurologists. It is included because a hand-written estimator that agrees with `statsmodels` to machine precision is the cheapest available evidence that the algebra is right.

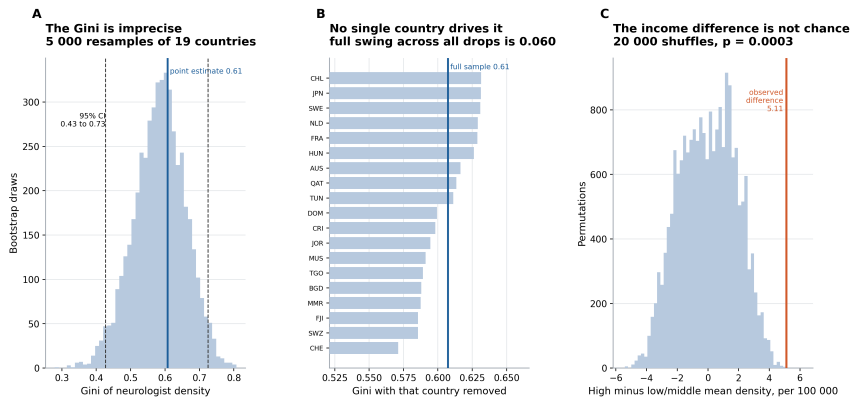
The interaction the 13-country sample cannot resolve, and the comparison it can



Panel A: each point is one of the 13 countries carrying both a burden-to-capacity ratio and a service-reach level. Both axes are logarithmic. Lines are ordinary least squares fits within each group; if capacity bought less alignment where services stay urban, the orange line would be flatter than the blue one. It is not, and the interaction coefficient is indistinguishable from zero. That is a statement about statistical power at $n = 13$, not evidence that no interaction exists. Panel B shows the comparison the larger service-reach sample does support: rural reach is far more common in high-income countries. Source: WHO Global Dementia Observatory (2017) and WHO Global Health Estimates. Cell counts are small throughout, so both panels are descriptive.

Figure 13: The interaction the 13-country sample cannot resolve, beside the comparison it can. Panel A: two nearly parallel fits, interaction coefficient -0.004 at $P = 0.99$. Panel B: rural reach is far more common in high-income countries. The title states what the figure shows, including that the sample cannot resolve the first question.

Robustness of the country-level results to their small sample



Panel A resamples the 19 countries with replacement 5 000 times and recomputes the Gini each time. The interval is wide because the sample is small, so the coefficient should be read as evidence of high inequality rather than as the precise value 0.61. Panel B recomputes the Gini dropping each country in turn, which checks that no single reporter carries the result; the full swing is 0.060. Panel C shuffles the income labels 20 000 times to build the null distribution for the difference in means directly, which avoids any normality assumption at $n = 8$ versus 7. All three use WHO Global Dementia Observatory neurologist density per 100 000 population (2017).

Figure 14: Robustness. Bootstrap distribution of the Gini, leave-one-out sensitivity, and the bounding exercise. Dropping any single country moves the Gini by at most 0.06, and burden would have to be overstated by a factor of 34 in low- and middle-income countries for the central result to reverse.

A.5 Omitted tables

Seven tables hold the numbers behind results stated in prose above and are not reproduced here: the Atlas gradients (t03), the regression output (t05), the interaction tests (t06), the regional comparison (t07), the paired-instrument correlations (t08), DHS coverage counts (t09), and the leave-one-out and bounding results (t10). All are committed as CSV in `output/tables/`.

Agentic Analysis

An AI assistant (Claude, Anthropic) was used throughout. Transcripts are in `docs/agentic/`. This section is excluded from the page limit.

My own role is quickly stated. The research questions, the choice of a share ratio over a difference, the decision to bin rather than regress, and the bounding exercise are mine; I executed and read every line of code, and traced every reported number back to a source table. The space below goes to the four episodes where the assistant was wrong, because the diagnoses transfer and the successes do not.

Rejected: a regression presented as a headline. The assistant proposed fitting the alignment ratio on capacity and income and reporting the slope as the paper’s main quantitative result, with a standard error and a p -value.

How it was wrong: the estimate is not identified in any useful sense on 19 observations, but the output does not look that way. A coefficient with a standard error carries an implicit claim to precision that the sample cannot support, and nothing in the fitted object flags this. I replaced it with binned comparisons and a 20,000-draw permutation test, which make a weaker claim honestly. The regression survives only in Fig. 12, as a check that two solvers agree.

Rejected: dropping the countries that break the plot. Eswatini and Fiji report zero neurologists, so the share ratio is undefined for them and they cannot be drawn on a logarithmic axis. The assistant proposed excluding them.

How it was wrong: the suggestion optimised for a clean figure over a correct sample. Those two countries are the most extreme instances of the phenomenon the paper is about, and their stroke DALY rates, 1,536 and 994 per 100,000, are among the highest in the sample. Dropping them would have removed the strongest evidence for the claim while improving the axis. They are reported separately in the text and named in the caption instead.

Failure: a figure title asserting a finding the data did not support. An earlier revision produced a panel titled “adding capacity does not help equally in every country,” describing an interaction whose coefficient was -0.004 at $P = 0.99$.

How it was wrong: the code ran, the figure rendered, and the title stated a substantive conclusion drawn from a null result on 13 observations. Nothing in the pipeline connects a title string to the significance of what it describes, so this class of error is invisible to testing. The panel now reads “the interaction the 13-country sample cannot resolve, beside the comparison it can,” which states what the figure shows including the part it cannot answer.

Failure: silent truncation in the PDF parser, and an over-correction. The Neurology Atlas wraps two long country names across lines. The first parser took only the first line, yielding “The former Yugoslav Republic of” as a country name.

How it was wrong, first attempt: the parse succeeded, produced 133 rows as expected, and the corrupted values sat in a column nobody reads closely. Row counts are the usual check and row counts were correct.

How it was wrong, second attempt: the assistant’s fix appended any short following line to the country name. This over-corrected, gluing contributor names onto countries and producing entries such as “Austria Grisold, Regina Katzenschlager.” The rule that held keys on the observation that a wrapped country name is cut

mid-phrase and ends in a function word, which no contributor name does. Eight assertions against values read off the page now guard the parse.

Failure: a data exposure I was not warned about. The DHS inventory the assistant built retained the authenticated download URLs in a committed column. Those URLs are tied to my approved DHS account.

How it was wrong: nothing downstream used the column, so its presence changed no result and produced no error. It would have been published to a public repository. `02_build_dhs_inventory.py` now discards the URLs at parse time, and the exclusion is stated in the script docstring so it cannot be reintroduced by someone rebuilding the file.

What this implies. All four share a shape: output that is internally consistent and externally wrong, produced with the same confidence whether right or not. None would have been caught by a test suite, because in each case the code did what it said. Each surfaced only when I asked what a quantity *meant* rather than whether the script ran. AI assistance moved the bottleneck from writing code to verifying that the code answered the intended question, and the second is the harder job.

Materials and Methods

All code, committed data and figures are at <https://github.com/alannap27/qss20-neuro-access>, organized as numbered scripts in `code/`, committed inputs in `data/`, and figures and tables in `output/`. Shared functions are in `code/utils.py` and figure layout rules in `code/figstyle.py`. No script contains a hardcoded absolute path. The WHO inputs are public and committed; the Neurology Atlas PDF and the DHS manifest are excluded, the first because it is a copyrighted publication and the second because its URLs are tied to an approved DHS account. Analyses used Python 3.10 with pandas, NumPy, SciPy, statsmodels and Matplotlib; versions are pinned in `requirements.txt`. Random seeds are fixed throughout.

References

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